

Q4. Explain with your own example the difference between entry-controlled and exit-controlled loops by writing a short code snippet for each (for/while vs do-while) and describing what happens if the loop condition is false at the start.

Ans:

1. Entry-Controlled Loop (for / while)

In an **entry-controlled loop**, the condition is checked **before** the loop body executes. If the condition is false at the beginning, the loop body is **not executed even once**.

2. Exit-Controlled Loop (do-while)

In an **exit-controlled loop**, the condition is checked **after** the loop body executes. Therefore, the loop body is **executed at least once**, even if the condition is false initially.